

CS-23334 FUNDAMENTALS OF DATA SCIENCE

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Experiment 7

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7. Experiment to understand Linear Regression for a given data set.

Aim:

To understand and evaluate Linear Regression Model for predicting the continuous dependent variable

Description:

Understand the Linear regression for the dataset given.

Algorithm:

Step 1: Define the Problem and Select Features

Step 2: Split the Dataset into Training and Testing Sets

Step 3: Train the Linear Regression Model

Step 4: Evaluate Model Performance Using Metrics

Step 5: Visualize Predictions and Residuals

About Dataset:

This dataset shows the relationship between years of professional experience and the corresponding annual salary for employees.

Code With Output:

```
import numpy as np
import pandas as pd
df=pd.read_csv(r'D:\REC 2nd Year\Data Science\Data Sets\Linear
Regression Dataset.csv')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  -
0   YearsExperience  30 non-null    float64
1   Salary          30 non-null    int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
df.dropna(inplace=True)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
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```

```
features=df.iloc[:, [0]].values
label=df.iloc[:, 1].values
```

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state=23)
```

```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
```

▼ LinearRegression

► Parameters

```
model.score(x_train,y_train)
```

```
0.9603182547438908
```

```
model.score(x_test,y_test)
```

```
0.9184170849214232
```

```
model.coef_
```

```
array([9281.30847068])
```

```
model.intercept_
```

```
np.float64(27166.736828912777)
```

```
import pickle
```

```
pickle.dump(model,open('SalaryPred.model','wb'))
```

```
model=pickle.load(open('SalaryPred.model','rb'))
```

```
yr_of_exp=float(input("Enter Years of Experience: "))
```

```
yr_of_exp_NP=np.array([[yr_of_exp]])
```

```
Salary=model.predict(yr_of_exp_NP)
```

```
Enter Years of Experience: 44
```

```
print("Estimated Salary for {} years of experience is {}:  
" .format(yr_of_exp,Salary))
```

```
Estimated Salary for 44.0 years of experience is [435544.30953887]:
```

Result:

Thus evaluating Linear Regression for a dataset is completed using a python program.